

FileEditSelectionViewGoRun

AI Lab Programs

AI_Program_1AI_Program_2AI_Program_3AI_Program_4XAI_Program_1(24BECS124).pdf

OPEN EDITORS

AI_Program_1AI_Program_2AI_Program_3X AI_Program_4AI_Program_1(24BECS124).pdf

AI LAB PROGRAMS

.dist.gitattributesAI_Program_1AI_Program_1(24BECS124).pdfAI_Program_2AI_Program_2(24BECS124).pdfAI_Program_3AI_Program_3(24BECS124).pdfAI_Program_4README.md

AI_Program_4 > ...

```
1 class RuleBasedSystem:
2     def __init__(self, facts, rules):
3         self.facts = set(facts)
4         self.rules = rules
5
6     def forward_chain(self):
7         iterations = 0
8
9         while True:
10             new_fact_added = False
11
12             for rule in self.rules:
13                 # Check if all conditions of the rule exist in current facts
14                 if all(cond in self.facts for cond in rule['if']):
15                     if rule['then'] not in self.facts:
16                         print(
17                             f"Rule Triggered: IF {rule['if']} "
18                             f"THEN Add {rule['then']}"
19                         )
20
21                     self.facts.add(rule['then'])
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Code + -

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_4"
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_human', 'Socrates_is_mortal', 'All_humans_are_mortal'}
PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

<main00

Geethajigaluru (54 seconds ago) Ln 60, Col 51 Spaces: 4 UTF-8 CRLF {} Python Sign In 3.13.14 (Microsoft Store) Go Live

FileEditSelectionViewGoRun

AI Lab Programs

AI Program 1AI Program 2AI Program 3AI Program 4 XAI Program_1(24BECS124).pdf

EXPLORER

OPEN EDITORS

AI Program 1AI Program 2AI Program 3AI Program 4 XAI Program_1(24BECS124).pdf

AI LAB PROGRAMS

.dist.gitattributesAI Program 1AI Program_1(24BECS124).pdfAI Program 2AI Program_2(24BECS124).pdfAI Program 3AI Program_3(24BECS124).pdfAI Program 4README.md

AI Program 4 > ...

3132 # Example Usage33 if __name__ == "__main__":3435 # Base facts36 initial_facts = [37 'Socrates_is_human',38 'All_humans_are_mortal'39]4041 # Production rules42 production_rules = [43 {44 'if': [45 'Socrates_is_human',46 'All_humans_are_mortal'47],48 'then': 'Socrates_is_mortal'49 }50]51

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_4"

Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_human', 'Socrates_is_mortal', 'All_humans_are_mortal'}
PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

main00

Geethajigaluru (54 seconds ago)Ln 60, Col 51Spaces: 4UTF-8CRLF{}PythonSign In3.13.14 (Microsoft Store)Go Live